

Protein-Ligand Binding Prediction

Machine Learning on the KIBA Dataset

1 000 proteins benchmark · Cold-Both evaluation · 5 models + CV

ESM-2 protein embeddings · Morgan + MACCS fingerprints · RDKit descriptors

Pipeline Overview



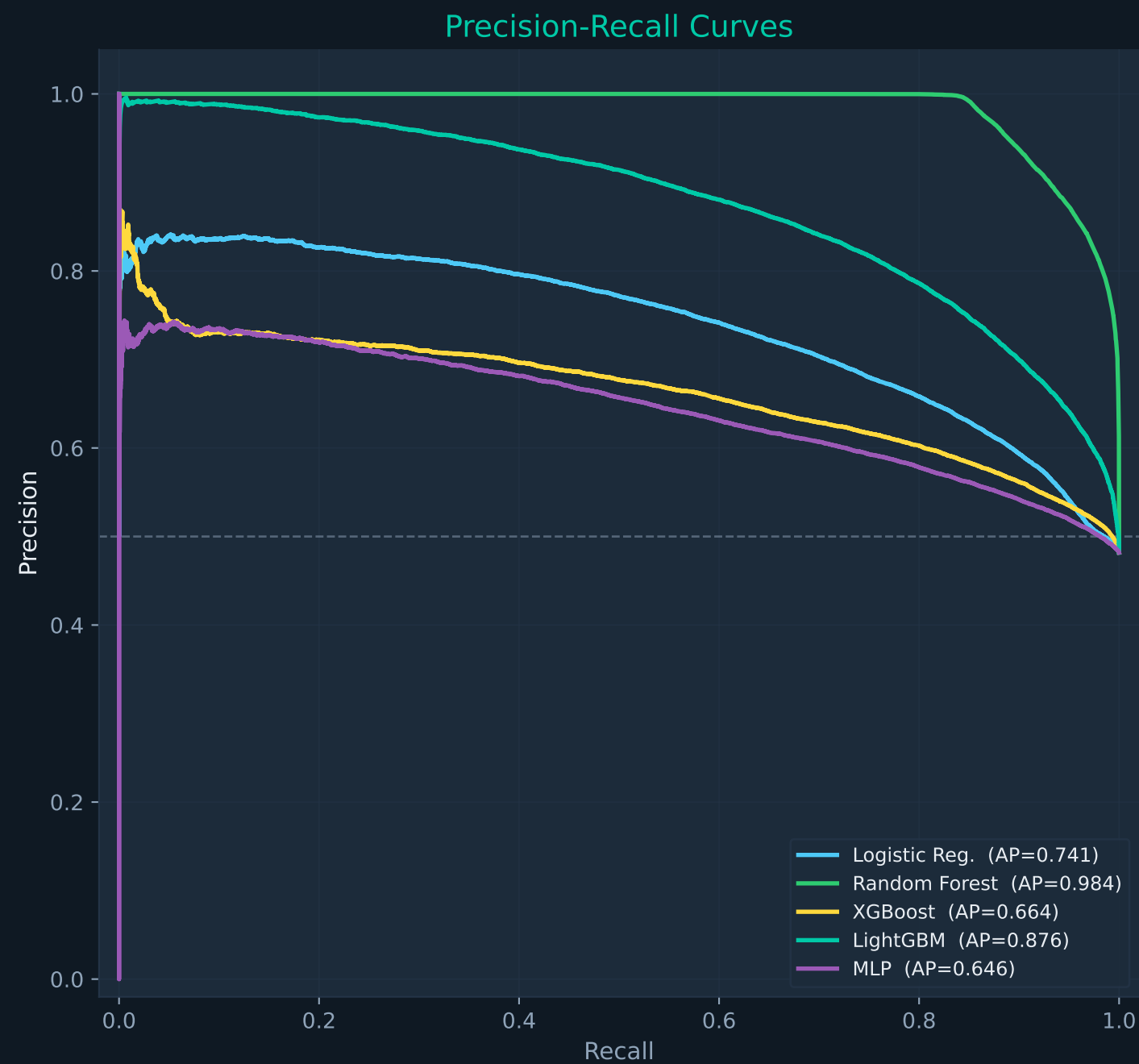
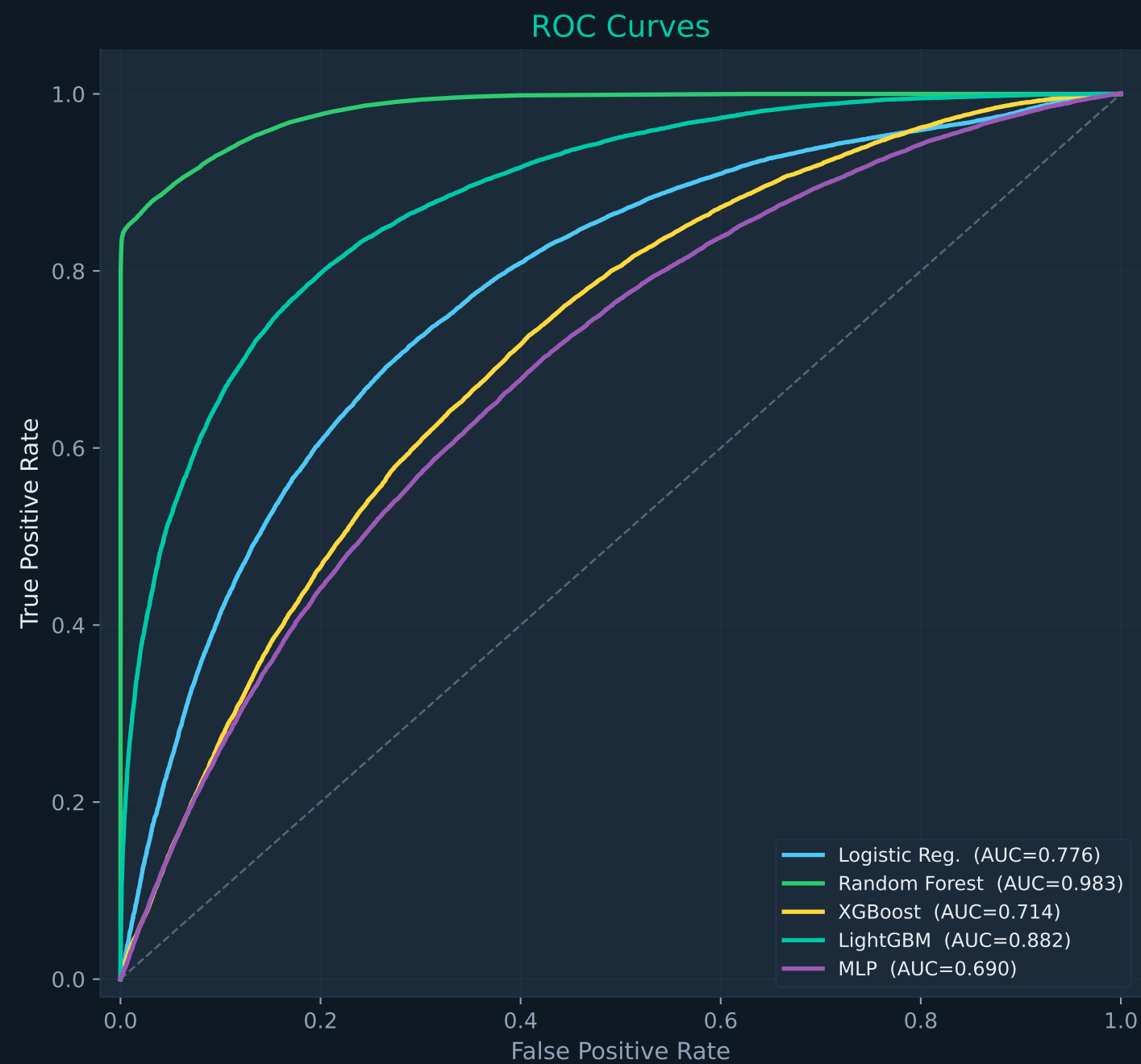
Model Comparison — Test Set (Cold-Both Split)

Unseen proteins AND unseen ligands in test · 1 000 proteins



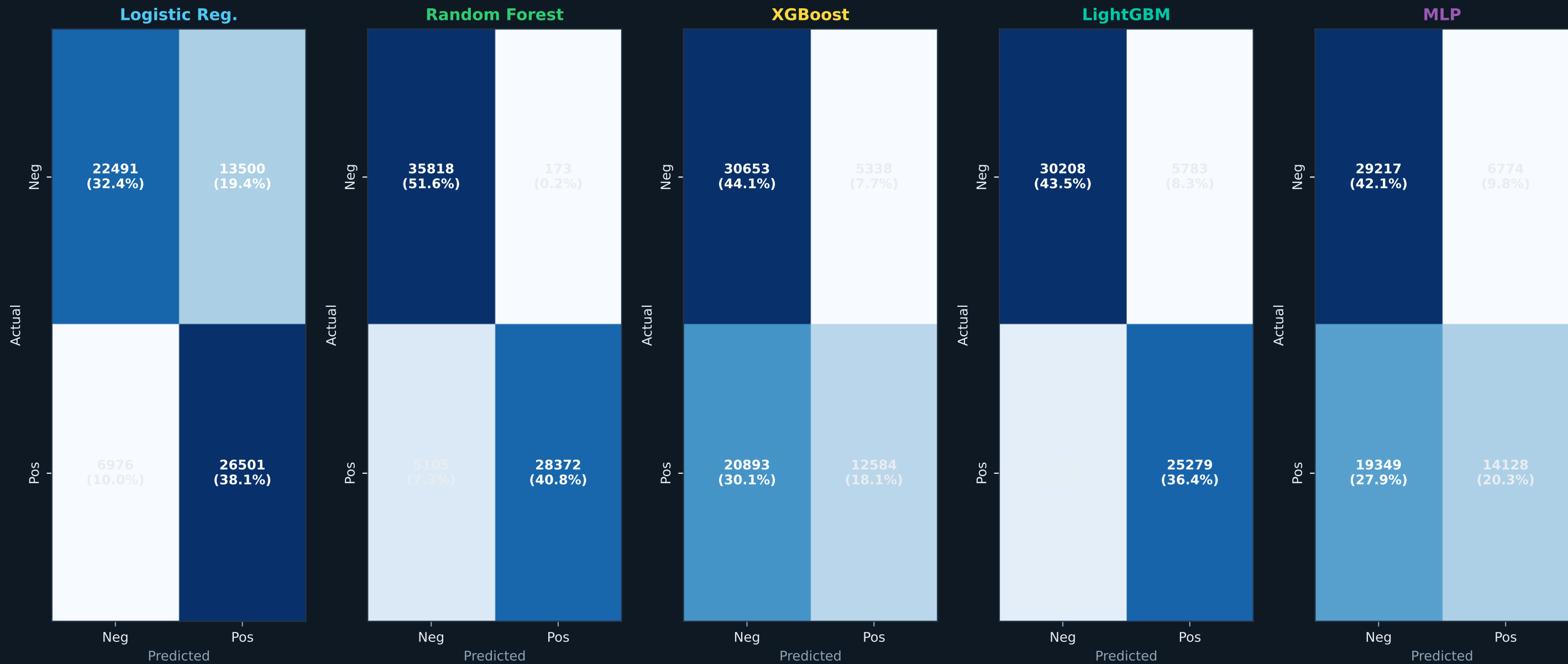
ROC and PR Curves — Cold-Both Test Set

Test proteins and test ligands are entirely unseen during training



Confusion Matrices — Cold-Both Test Set

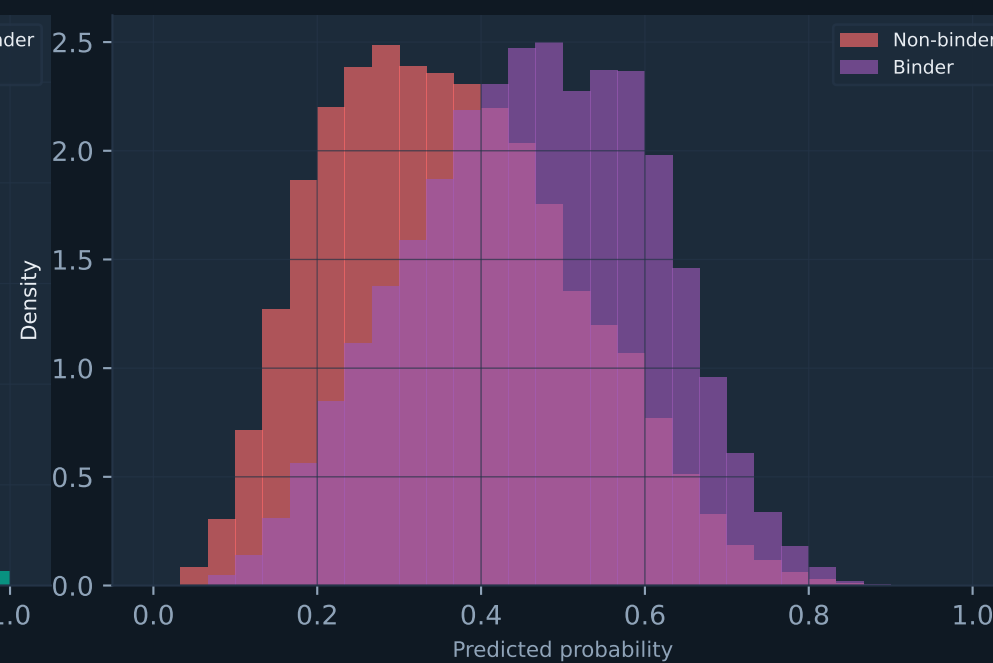
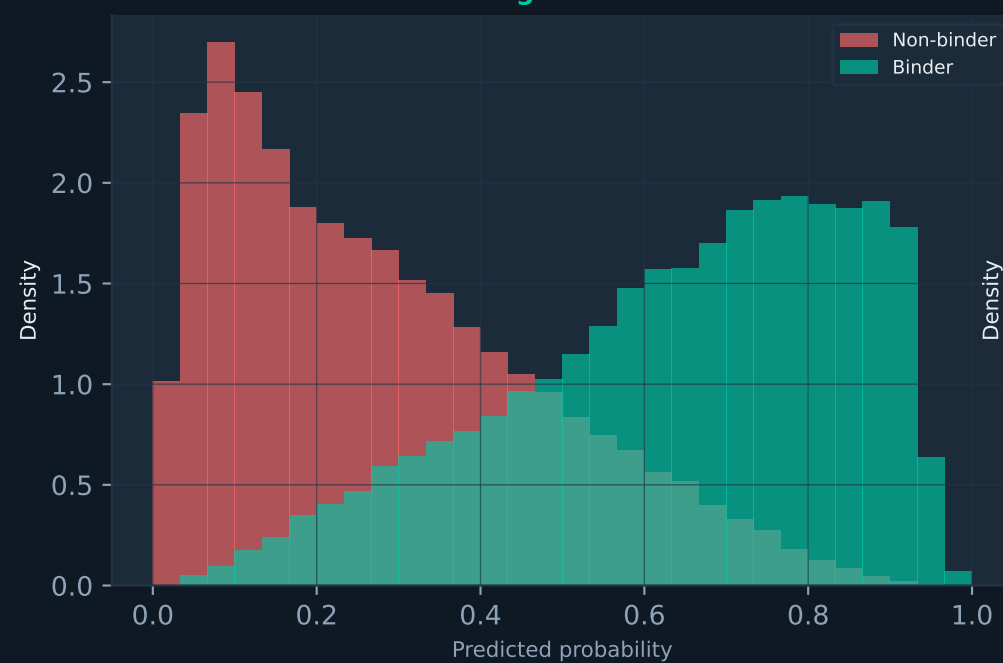
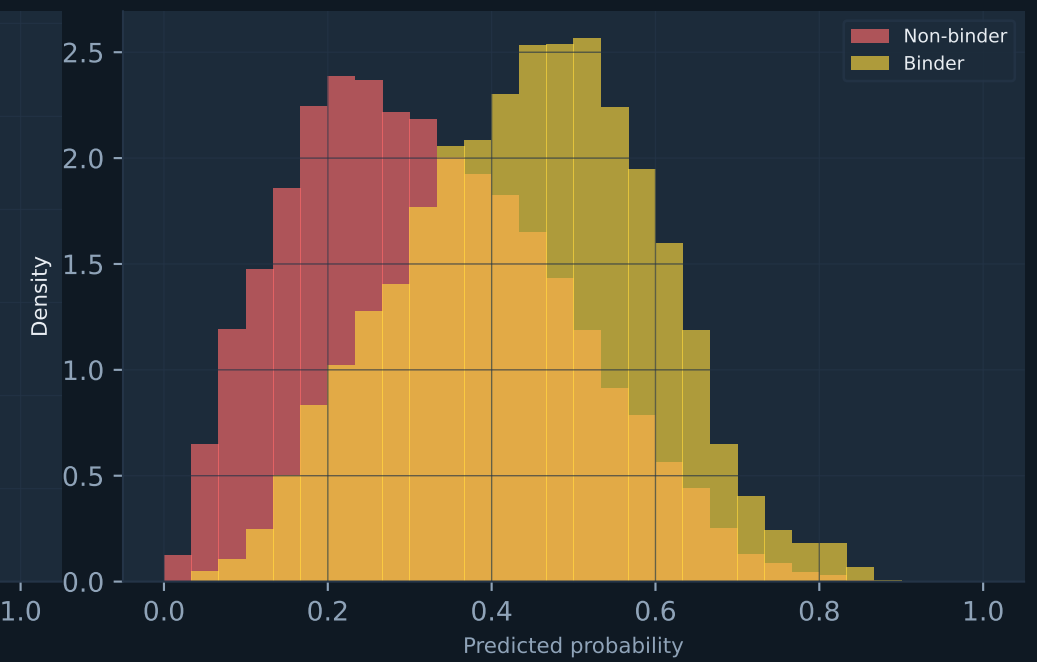
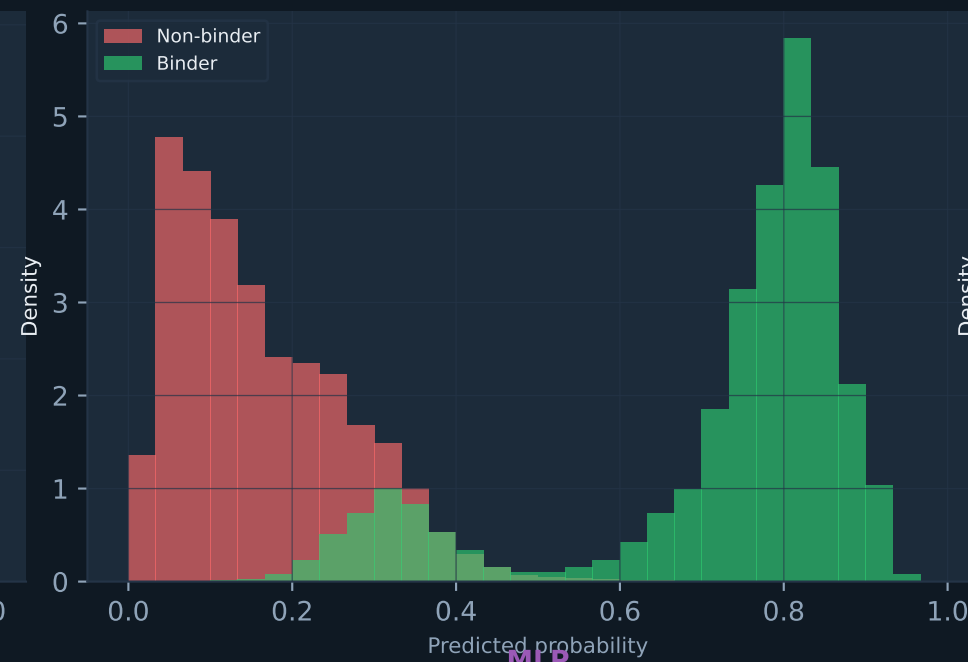
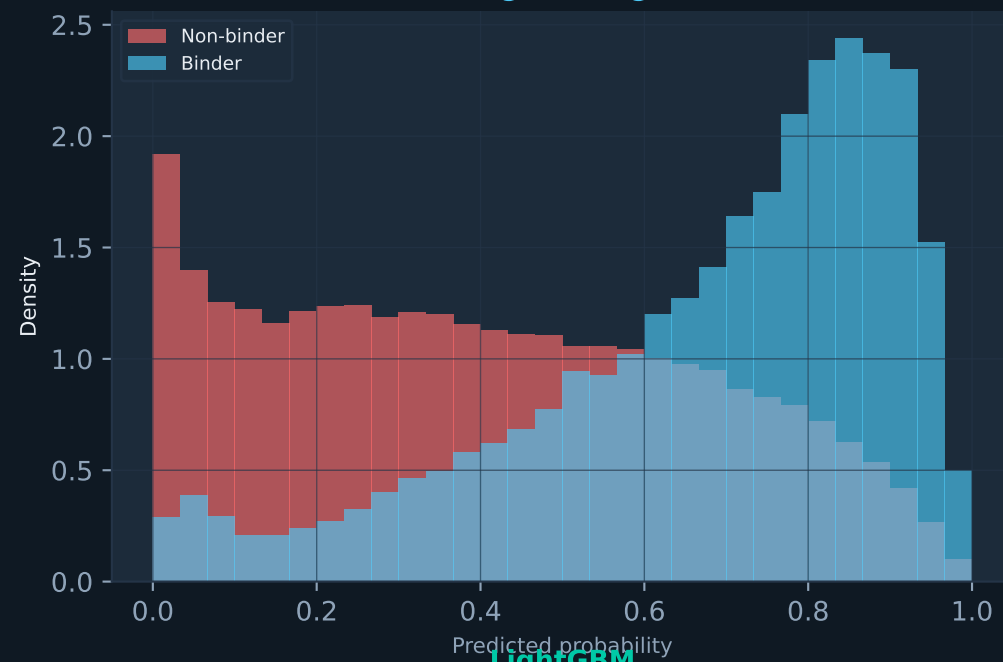
Evaluation samples from the 1000-protein cold_both run



Predicted Score Distributions

Random Forest

XGBoost



5-Fold Protein-Aware Cross-Validation

StratifiedGroupKFold — no protein appears in both train and validation folds



CV Summary		
Model	ROC-AUC	PR-AUC
Logistic Reg.	0.628±0.028	0.587
Random Forest	0.697±0.004	0.651
XGBoost	0.704±0.016	0.656
LightGBM	0.709±0.018	0.663

CV is consistent with test set —
models generalise well to unseen proteins.

Results Summary — All Models

Cold-Both split · 1 000 proteins · 5-fold protein-aware CV

Model	ROC-AUC	PR-AUC	F1	BEDROC	EF@1%	CV ROC-AUC	CV ± Std
Logistic Reg.	0.6061	0.5628	0.4888	0.8269	1.7168	0.6284	0.0279
Random Forest	0.7007	0.6598	0.1678	0.9999	2.0751	0.6972	0.0041
XGBoost	0.7137	0.6639	0.4897	0.7315	1.6929	0.7038	0.0157
LightGBM	0.7151	0.6757	0.4120	0.9712	2.0572	0.7090	0.0176
MLP	0.6897	0.6463	0.5196	0.7239	1.4869	nan	nan

Limitations & Next Steps

Current Limitations

- Negatives are property-matched decoys, not true
- measured non-binders — may inflate performance
- 1 000 proteins subset used for this benchmark
- results may vary at full dataset scale
- KIBA scores are partially estimated (not all measured)
- Ligand features are fingerprint-based — no 3-D geometry

Next Steps

- Graph Neural Networks on molecular graphs (GNN-DTI)
- Structure-aware models: AlphaFold2 binding site features
- Multi-task learning across kinase families
- Active learning loop: prioritise wet-lab validation hits
- Larger benchmark: full KIBA + BindingDB + ChEMBL